

JAMES RUSE AHS
MATH. EXT 1 TRIAL, 2008

Question 1. **Marks**

- (a) Find $\lim_{x \rightarrow 0} \frac{3x}{\tan 5x}$. 2
- (b) Find the obtuse angle between the lines $x - y - 1 = 0$ and $2x + y - 1 = 0$. 2
- (c) Find the general solution to $\sin \theta = \frac{\sqrt{3}}{2}$. 2
- (d) When the polynomial function $f(x)$ is divided by $x^2 - 16$, the remainder is $3x - 1$. What is the remainder when $f(x)$ is divided by $x - 4$? 2
- (e) Solve for x : $\frac{1-2x}{1+x} \geq 1$. 3
- (f) Find a primitive of $\frac{1}{\sqrt{x^2 - 9}}$. 1

Question 2. **[START A NEW PAGE]**

- (a) Given the function $g(x) = \sqrt{x+2}$ and that $g^{-1}(x)$ is the inverse function of $g(x)$, find $g^{-1}(5)$. 2
- (b) (i) Show that: $\frac{2 \tan x}{1 + \tan^2 x} = \sin 2x$. 1
- (ii) Hence, or otherwise, find $\int_0^{\frac{\pi}{4}} \frac{\tan x}{1 + \tan^2 x} dx$. 2
- (c) Using the substitution $u = \sqrt{1+x}$, evaluate $\int_0^3 \frac{5x^2 + 10x}{\sqrt{1+x}} dx$. 4
- (d) Sketch the graph of the curve: $y = 2 \cos^{-1}(x) - 1$, showing all essential information. 3

Question 3. **[START A NEW PAGE]** **Marks**

- (a) Find the exact value of $\tan\left(2 \cos^{-1} \frac{12}{13}\right)$. 2
- (b) Let point $P(4p, 2p^2)$ be an arbitrary point on the parabola $x^2 = 8y$ with parameter p .
- (i) Show that the equation of the tangent at P is $y = px - 2p^2$. 1
- (ii) The tangent intersects the y -axis at C .
The point Q divides CP , internally, in the ratio $1:3$.
Find the locus of all the Q points as parameter p varies. 3
- (c) The velocity $v \text{ ms}^{-1}$ of a particle moving in a straight line at position x at time t seconds is given by: $v = x^3 - x$.
Find the acceleration of the particle at any position. 2
- (d) The numbers 1447, 1005 and 1231 all have something in common.
Each is a four-digit number beginning with 1 that has exactly two identical digits.
How many such four-digit numbers exist? 2
- (e) Find $\int \cos^2\left(\frac{x}{2}\right) dx$. 2

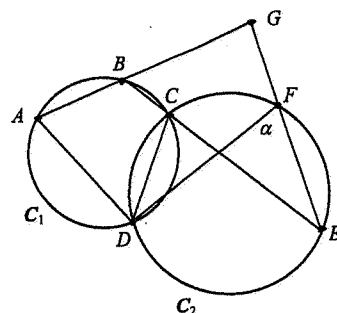
Question 4.

[START A NEW PAGE]

Marks

- (a) Find the term independent of x in the expansion of $\left(2x^2 - \frac{3}{x}\right)^9$. 2

(b) 3



Two circles C_1 and C_2 intersect at C and D .
 BC produced meets circle C_2 at E .
 AB produced meets EF produced at G .
 Let $\angle DFE = \alpha$.

Copy or trace the diagram onto your writing booklet and prove that $ADFG$ is a cyclic quadrilateral.

- (c) A bag contains eleven balls, numbered 1, 2, 3, ... and 11. If six balls are drawn simultaneously at random,
- (i) How many ways can the sum of the numbers on the balls drawn be odd? 2
- (ii) What is the probability that the sum of the numbers on the balls drawn is odd? 1
- (d) When Farmer Browne retired he decided to invest \$2 000 in a fund which paid interest of 8% *pa*, compounded annually. From this fund he decided to donate a yearly prize of \$200 to be awarded to the Dux of Agriculture in Year 12. The prize money being withdrawn from this fund after the year's interest had been added.
- (i) Show that the balance $\$B_n$ remaining after n prizes have been awarded will be: $B_n = 500(5 - 1.08^n)$ 3
- (ii) Calculate the number of years that the \$200 prize can be awarded. 1

Question 5.

[START A NEW PAGE]

Marks

- (a) Considering the expansion:
 $(9 + 5x)^{29} = p_0 + p_1x + p_2x^2 + \dots + p_kx^k + \dots + p_nx^n$.
- (i) Use the Binomial theorem to write the expression for p_k . 1
- (ii) Show that: $\frac{p_{k+1}}{p_k} = \frac{5(29-k)}{9(k+1)}$. 2
- (ii) Hence, or otherwise, find the largest coefficient in the expansion. 2
 [you may leave your answer in the form: $\binom{29}{r} 3^a 5^b$].
- (b) An ice cube tray is filled with water which is at a temperature of 20°C and placed in a freezer that is at a constant temperature of -15°C . The cooling rate of the water is proportional to the difference between the temperature of the water $W^\circ\text{C}$, so that W satisfies the rate equation:
 $\frac{dW}{dt} = -k(W + 15)$, where k is the rate constant of proportionality.
- (i) Show that: $\frac{d}{dt}(We^{kt}) = -15ke^{kt}$. 2
- (ii) Hence, show that: $W = 35e^{-kt} - 15$. 2
- (iii) After 5 minutes in the freezer, the temperature of the water cubes is 6°C .
1. Find the rate of cooling at this time (correct to 1 decimal place) 2
2. Find the time for the water cubes to reach -10°C (correct to the nearest minute). 1

Question 6.

[START A NEW PAGE]

Marks

- (a) A ball is projected from a point O on horizontal ground in a room of length $2R$ metres with an initial speed of $U \text{ ms}^{-1}$ at an angle of projection of α . There is no air resistance and the acceleration due to gravity is $g \text{ ms}^{-2}$.
- (i) Assuming after t seconds the ball's horizontal distance x metres, is given by: $x = Ut \cos \alpha$, and the vertical component of motion is $\ddot{y} = -g$, show that the vertical displacement y of the ball is given by:
- $$y = Ut \sin \alpha - \frac{1}{2}gt^2.$$
- (ii) Hence show that the range R metres for this ball is given by:
- $$R = \frac{U^2 \sin 2\alpha}{g}.$$
- (iii) Suppose that the room has a height of 3.5 metres and the angle of projection is fixed for $0 < \alpha < \frac{\pi}{2}$ but the speed of projection U varies. Prove that:
- (α) the maximum range will occur when $U^2 = 7g \operatorname{cosec}^2 \alpha$.
- (β) the maximum range would be $14 \cot \alpha$.

- (b) Given the polynomial function:

$$f_n(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!} + \dots + \frac{x(x+1)\dots(x+n-1)}{n!}, \text{ for } n = 1, 2, 3, \dots$$

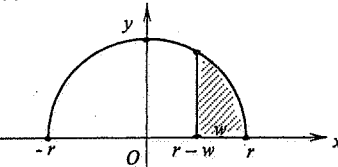
where for $n = 1$: $f_1(x) = 1 + \frac{x}{1!} = x + 1$ which has a zero at -1 .

- (i) Show that for $n = 2$: $f_2(x) = \frac{1}{2!}(x+1)(x+2)$ and state the zeros of $f_2(x)$.
- (ii) Hence **complete** the proof by mathematical induction that the zeros of the polynomial function $f_n(x)$ are $-1, -2, -3, \dots$ and $-n$ for $n = 1, 2, 3, \dots$, that is
- prove that: $f_n(x) = \frac{1}{n!}(x+1)(x+2)(x+3)\dots(x+n)$, for $n = 1, 2, 3, \dots$

Question 7.

[START A NEW PAGE]

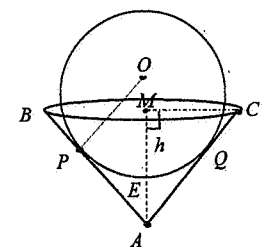
Marks

- (a) Given the semi-circle equation: $y = \sqrt{r^2 - x^2}$,
- 
- The shaded area of thickness w is rotated about the x -axis to form the volume of a 'cap'.

Show that the volume of the solid of revolution V is given by:

$$V = \frac{\pi}{3}(3r - w)w^2.$$

- (b) An inverted cone ABC of height H units with a base radius of R units is filled with water. A sphere of radius r units is inserted into the inverted cone so as to touch the inner walls of the cone at P & Q to a depth of h units, as shown below.



Not to scale

Given:

$MB = MC = R$, $MA = H$, $AC = L$,
 $OP = r$ and $ME = h$.

- (i) Show that: $r = \frac{(H-h)R}{L-R}$, where $L = \sqrt{H^2 + R^2}$.
- (ii) Hence show that the volume of water V cubic units displaced by the sphere is given by:
- $$V = \frac{\pi}{3(L-R)}[3RHH^2 - (L+2R)h^3].$$
- (iii) Hence, or otherwise find the radius of the sphere that displaced the maximum volume of water under the above conditions.
- (c) (i) Write down the binomial expansion of $(1-x)^{2n}$ in ascending powers of x .
- (ii) Hence show that:
- $$\binom{2n}{1} + 3\binom{2n}{3} + \dots + (2n-1)\binom{2n}{2n-1} = 2\binom{2n}{2} + 4\binom{2n}{4} + \dots + 2n\binom{2n}{2n}.$$

THE END ☺ ☹ ☹ ☹

MATHEMATICS Extension 1 : Question 1

Suggested Solutions

Marks

Marker's Comments

Q1(a) $\lim_{x \rightarrow 0} \frac{3x}{\tan 5x} = \lim_{x \rightarrow 0} \frac{3 \times (5x)}{\tan(5x) \times 5}$
 $= \frac{3}{5} \lim_{x \rightarrow 0} \frac{5x}{\tan 5x}$
 $= \frac{3}{5} \times 1$
 $= \frac{3}{5}$

[2]

(b) $x - y - 1 = 0$ $m_1 = 1$
 $2x + y - 1 = 0$ $m_2 = -2$
 $\tan \theta = \frac{-2 - 1}{1 + (-2)(1)} = \frac{-3}{1 - 2} = 3$
 $\therefore \tan \theta = 3$
 $\therefore \text{obtuse angle} = 180^\circ - \tan^{-1} 3$
 $= 108^\circ 26'$

[2]

(c) $\sin \theta = \frac{\sqrt{3}}{2}$
 $\theta = n\pi + (-1)^n \sin^{-1} \frac{\sqrt{3}}{2}$
 $\theta = n\pi + (-1)^n \frac{\pi}{3}$ where $n \in \mathbb{Z}$

[2]

(d) $f(x) = (x^2 - 16)Q(x) + 3x - 1$
 $\text{Rem} = f(4) = 0 + 3 \times 4 - 1$
 $= 11$

[2]

(e) $\frac{1-2x}{1+x} \geq 1$
 $1-2x - 1(1+x) \geq 0$
 $\frac{1+x}{1+x} - \frac{3x}{1+x} \geq 0$
 $\frac{3x}{1+x} \leq 0$
 $\text{Now } x \neq -1$
 $\therefore 3x(1+x) \leq 0$
 $\Rightarrow -1 < x \leq 0$

[3]

(f) Primitive $\ln[x + \sqrt{x^2 - 9}] (+C)$

[1]

or $\tan^{-1}(3)$

or $\theta = \begin{cases} \frac{\pi}{3} + 2n\pi \\ \pi - \frac{\pi}{3} + 2n\pi \end{cases}$
 $\text{Acc } \theta = 180^\circ n + (-1)^n 60^\circ$

MATHEMATICS Extension 1 : Question 2

Suggested Solutions

Marks

Marker's Comments

Q2(a) $g(x) = \sqrt{x+2}$
 $g^{-1}(5)$ is $g(x) = 5$
 $5 = \sqrt{x+2}$
 $\therefore x = 23$

[2]

(b) (i) $\frac{2 + \tan x}{1 + \tan^2 x} = \frac{2 \sin x}{\cos x} = \frac{2 \sin x \cos x}{\cos^2 x}$
 $= 2 \sin x \cos x$
 $= \sin 2x$ check

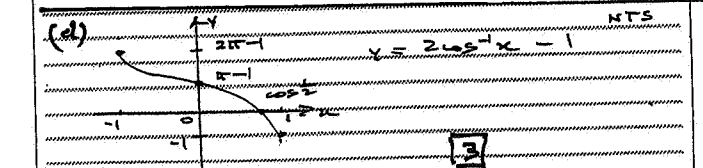
[1]

(ii) $\int_0^{\pi/4} \frac{\tan x}{1 + \tan^2 x} dx = \frac{1}{2} \int_0^{\pi/4} \sin 2x dx$
 $= -\frac{1}{4} [\cos 2x]_0^{\pi/4}$
 $= -\frac{1}{4} [\cos \frac{\pi}{2} - \cos 0]$
 $= -\frac{1}{4} [0 - 1]$
 $= \frac{1}{4}$

[2]

(c) $I = \int_0^3 \frac{5x^2 + 10x}{1+x} dx$
 $u = 1+x$
 $u^2 = 1+x$
 $2u du = dx$
 $\therefore I = \int_1^4 \frac{(u-1)^2 + 2(u-1) \times 2u}{u} du$
 $= \int_1^4 (u^2 - 2u + 1 + 2u - 2) du$
 $= \int_1^4 (u^2 - 1) du$
 $= 10 \left[\frac{1}{3} u^3 - u \right]_1^4 = 10 \left[\left(\frac{64}{3} - 4 \right) - \left(\frac{1}{3} - 1 \right) \right]$
 $= 10 \left(\frac{58}{3} - 1 \right) = 52$

[4]



[3]

$g^{-1}(x) = x^2 - 2$

1 For $x = \cos \frac{\pi}{2} = 0.88$
 1 For $2\pi - 1$ and -1
 $\frac{1}{2}$ For $\pi - 1$
 $\frac{1}{2}$ For shape

MATHEMATICS Extension 1 : Question 3

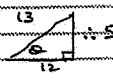
Suggested Solutions

Marks

Marker's Comments

Q3(a) $\tan(2\cos^{-1}\frac{12}{13})$

Let $\theta = \cos^{-1}\frac{12}{13} \Rightarrow \cos\theta = \frac{12}{13}$



$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2 \times \frac{5}{12}}{1-\frac{25}{169}} = \frac{2 \times 5 \times 12}{169-25} = \frac{120}{144} = \frac{5}{6}$

1 For $\cos\theta = \frac{12}{13}$
 1 For $\tan\theta = \frac{5}{12}$
 1 For $\frac{2 \times \frac{5}{12}}{1-\frac{25}{169}}$
 or $\frac{120}{144}$

(b) (i) $x^2 = 8y$
 $\therefore y = \frac{x^2}{8}$ P(4p, 2p²)

$\frac{dy}{dx} = \frac{2x}{8} = \frac{x}{4}$

Gradient of tangent at P: $m_T = \frac{4p}{4} = p$

Equ. of tangent at P: $y - 2p^2 = p(x - 4p)$
 $y - 2p^2 = px - 4p^2$
 $\therefore y = px - 2p^2$

1 For getting to ✓

(ii) $C = (0, -2p^2)$

For Q (0, -2p²) 1:2 P(4p, 2p²)

Q = $(\frac{4p+0}{4}, \frac{2p^2-6p^2}{4}) = (p, -p^2)$

Let Q(x, y) be the general point on the required locus

$\therefore x = p$ — (1)

$y = -p^2$ — (2)

(1) $\Rightarrow p = x$ in (2) $y = -(x)^2$

$\therefore$ locus of Q $x^2 = -y$

(c) $v = x^3 - x$

$\frac{dv}{dx} = \frac{d}{dx}(x^3 - x) = 3x^2 - 1$

$\frac{dv}{dx} = (x^2 - x)(3x^2 - 1)$

(d) For two 1s 1 N^o of ways = $3 \times 98 = 294$
 For not having two 1s 1 N^o of ways = $9 \times 3 \times 8 = 216$
 or $4 \times 2 \times 1 \times 9 \times 8 = 432$ TOTAL $\frac{294+216}{432}$

(e) $I = \int \cos^2 \frac{x}{2} dx = \frac{1}{2} \int (1 + \cos x) dx$
 $= \frac{1}{2} [x + \sin x] + C$

1 For $\cos^2 A = \frac{1}{2}(1 + \cos 2A)$
 or equiv.

MATHEMATICS Extension 1 : Question 4

Suggested Solutions

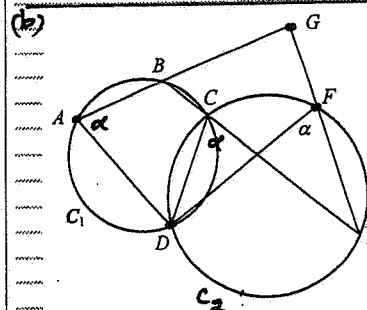
Marks

Marker's Comments

Q4(a) $(2x^3 - \frac{3}{x})^9$

General term $T_{r+1} = {}^9C_r (2x^3)^{9-r} (\frac{-3}{x})^r = A x^0$
 $\therefore {}^9C_r 2^{9-r} (-3)^r x^{27-3r} = A x^0$
 $\Rightarrow 27-3r = 0$
 $r = 9$

$\therefore$ Term is the seventh / $T_7 = {}^9C_6 2^3 (-3)^3 = -489888$



1. $\angle DCE = \alpha$ (Angles in same segment standing on arc DE are equal)
2. $\angle DAB = \alpha$ (Exterior angle of cyclic Quad ABC equals interior opposite angle)
3. $\angle AFE = \angle DAB = \alpha$
 $\therefore$ AGED is a cyclic Quad (Exterior angle equals interior opposite angle [converse])

(c) (i)

N^o of ways = $1000 + 3000 + 5000$
 $= {}^6C_1 \times 5^1 + {}^6C_2 \times 5^2 + {}^6C_3 \times 5^3$
 $= 6 + 20 \times 10 + 6 \times 5$
 $= 236$

Note: $0+0=E$
 $E+E=E$
 Need odd N^o of Ds for sum to be 01

(ii)

$P(E) = \frac{236}{462} = \frac{118}{231}$

(d) Let $P = 2000$ Int rate = 0.08, n is ...
 $A = 1.08$ $M = 200$

(i) After 1st prize:

$B_1 = P \times A - 200$

After 2nd prize awarded:

$B_2 = B_1 \times A - 200 = (PA - 200)A - 200$
 $= PA^2 - 200(1+A)$

After 3rd:

$B_3 = B_2 \times A - 200 = (PA^2 - 200(1+A))A - 200$
 $= PA^3 - 200(1+A+A^2)$

$\therefore$ After nth: $B_n = PA^n - 200(1+A+A^2+\dots+A^{n-1})$
 $= PA^n - 200 \frac{A^n - 1}{A - 1}$
 $= 2000A^n - \frac{200(A^n - 1)}{0.08}$
 $= 2000A^n - 2500(A^n - 1)$
 $= -500A^n + 2500 = 500[5 - 1.08^n]$ ned.

Set $B_n = 0$
 $\Rightarrow 1.08^n = 5$
 $n = \frac{\log 5}{\log 1.08} = 20.912$
 $\therefore$ N^o of years is 21

Marker's Comments

MATHEMATICS Extension 1 : Question 6		
Suggested Solutions	Marks	Marker's Comments
Q6(a)(i)  $t=0 \begin{cases} x=0 \\ y=0 \end{cases}$ $\dot{y} = -g$ $\dot{y} = \int -g dt$ $y = -gt + C$ but $t=0 \dot{y} = U \sin \alpha$ $\therefore C = U \sin \alpha$ $\dot{y} = U \sin \alpha - gt$ $y = \int (U \sin \alpha - gt) dt$ $y = Ut \sin \alpha - \frac{1}{2}gt^2 + D$ $t=0 \ y=0 \therefore D=0$ $\Rightarrow y = Ut \sin \alpha - \frac{1}{2}gt^2$	1	
(ii) For the range: $y=0$ $\therefore t(U \sin \alpha - \frac{1}{2}gt) = 0$ $\therefore t=0$ or $t = \frac{2U \sin \alpha}{g}$ $\therefore R = x = U \cdot \frac{2U \sin \alpha}{g} \cdot \cos \alpha = \frac{U^2 \sin 2\alpha}{g}$	1	
(iii) (a) Max height is 3.5m when $t = \frac{1}{2} \times \frac{2U \sin \alpha}{g} = \frac{U \sin \alpha}{g}$ $\therefore 3.5 = U \cdot \frac{U \sin \alpha}{g} \cdot \sin \alpha - \frac{1}{2}g \times \left(\frac{U \sin \alpha}{g}\right)^2$ $= \frac{U^2 \sin^2 \alpha}{g} - \frac{U^2 \sin^2 \alpha}{2g}$ $3.5 = \frac{U^2 \sin^2 \alpha}{2g}$ $\therefore U^2 = \frac{3.5 \times 2g}{\sin^2 \alpha} = \frac{7g \cos^2 \alpha}{\sin^2 \alpha}$	1	or $\dot{y} = U \sin \alpha - gt = 0$
(b) Max R will then be $R = \frac{U^2 \sin 2\alpha}{g}$ $= \frac{7g}{\sin^2 \alpha} \cdot \frac{2 \sin \alpha \cos \alpha}{g}$ $= \frac{14 \cos \alpha}{\sin \alpha}$ $\therefore \text{max } R = 14 \cot \alpha$	1	For subst. (iii) (a) into (ii) and showing how $14 \cot \alpha$

MATHEMATICS Extension 1 : Question 6...

Suggested Solutions

Marks

Marker's Comments

Q6(b)(i) $f_2(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!}$
 $= 2 + 2x + \frac{x(x+1)}{2} = \frac{2+2x+x^2+x^2+x}{2}$
 $= \frac{x^2+3x+2}{2} \checkmark$
 $= \frac{1}{2}(x+1)(x+2)$ (2)

and the zeros are -1 and -2

For getting to $\frac{x^2+3x+2}{2}$

(ii) Let $P(n)$ be the proposition that:

$f_n(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!} + \dots + \frac{x(x+1)\dots(x+n-1)}{n!} = \frac{1}{n!}(x+1)(x+2)\dots(x+n)$

Now $P(1)$ was given

$P(2)$ was shown true in part (i)

* Assume $P(n)$ is true for some integer k

i.e. $f_k(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!} + \dots + \frac{x(x+1)\dots(x+k-1)}{k!} = \frac{1}{k!}(x+1)(x+2)\dots(x+k)$ (*)

RTP: $P(k+1)$ is true

i.e. $f_{k+1}(x) = \frac{1}{(k+1)!}(x+1)(x+2)\dots(x+k+1)$

PROOF: For $P(k+1)$

$f_{k+1}(x) = 1 + \frac{x}{1!} + \frac{x(x+1)}{2!} + \dots + \frac{x(x+1)\dots(x+k-1)}{k!} + \frac{x(x+1)\dots(x+k)}{(k+1)!}$
 $= \frac{1}{k!}(x+1)(x+2)\dots(x+k) + \frac{x(x+1)\dots(x+k)}{(k+1)!}$ using assumption (*)

For using / substituting assumption

$= \frac{(x+1)(x+2)\dots(x+k)}{k!} \left\{ 1 + \frac{x}{k+1} \right\}$

$= \frac{(x+1)(x+2)\dots(x+k)}{k!} \left\{ \frac{k+1+x}{k+1} \right\}$

$= \frac{1}{(k+1)!}(x+1)(x+2)\dots(x+k+1)$

$\therefore P(k+1)$ is true (3)

* $\therefore$ by the PMI $P(n)$ is true for $n=1, 2, 3, \dots$

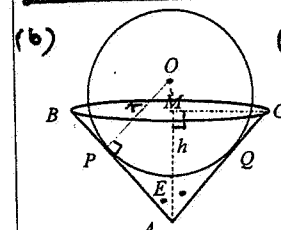
MATHEMATICS Extension 1 : Question 7...

Suggested Solutions

Marks

Marker's Comments

(a) $V = \pi \int_{r-w}^r (r^2 - x^2) dx$ (2)
 $= \pi \left[r^2x - \frac{1}{3}x^3 \right]_{r-w}^r$
 $= \pi \left[(r^3 - \frac{1}{3}r^3) - (r^2(r-w) - \frac{1}{3}(r-w)^3) \right]$
 $= \pi \left[\frac{2}{3}r^3 - (r-w)(3r^2 - (r-w)^2) \right]$
 $= \pi \left[\frac{2}{3}r^3 - (r-w)(3r^2 - r^2 + 2rw - w^2) \right]$
 $= \pi \left[\frac{2}{3}r^3 - (r-w)(2r^2 + 2rw - w^2) \right]$
 $= \pi \left[\frac{2}{3}r^3 - (2r^3 + 2r^2w - rw^2 - 2r^2w - 2rw^2 + w^3) \right]$
 $= \pi \left[\frac{2}{3}r^3 - 2r^3 + 2r^2w - rw^2 + 2r^2w - 2rw^2 + w^3 \right]$
 $= \pi \left[-\frac{4}{3}r^3 + 4r^2w - 3rw^2 + w^3 \right]$



(b) (i) $\Delta OPA \sim \Delta CMA$ (equiangular)
 $\frac{r}{R} = \frac{OA}{AC}$ (corresponding sides in similar Δ s are in the same ratio)
 $r = \frac{h}{L} (r-h)$
 $rL = h(r-h)$
 $r(L-h) = h^2$
 $\therefore r = \frac{h^2}{L-h}$ (2)

(ii) using (a) where $h=w$, $r = \frac{h^2}{L-h}$
 $\therefore V = \pi \int_{r-w}^r (r^2 - x^2) dx$
 $= \pi \left[r^2x - \frac{1}{3}x^3 \right]_{r-w}^r$
 $= \pi \left[\frac{2}{3}r^3 - (r-w)(3r^2 - (r-w)^2) \right]$
 $= \pi \left[\frac{2}{3}r^3 - (r-w)(2r^2 + 2rw - w^2) \right]$
 $= \pi \left[\frac{2}{3}r^3 - 2r^3 + 2r^2w - rw^2 + 2r^2w - 2rw^2 + w^3 \right]$
 $= \pi \left[-\frac{4}{3}r^3 + 4r^2w - 3rw^2 + w^3 \right]$

For subst and simplifying to

(iii) $\frac{dV}{dh} = \pi \left[6r^2h - 3(L+2r)h^2 \right]$
 $= \pi \left[2r^2h - (L+2r)h^2 \right]$

For possible max/min values of V to occur $\frac{dV}{dh} = 0$

$\therefore h(2r^2 - (L+2r)h) = 0$

(4) $\therefore h=0$ or $h = \frac{2r^2}{L+2r}$
 but $h \neq 0$

TEST: $\frac{d^2V}{dh^2} = \pi \left[2r^2 - 2(L+2r)h \right]$
 at $h = \frac{2r^2}{L+2r}$ $\frac{d^2V}{dh^2} = \pi \left[2r^2 - 4r^2 \right] = -2r^2 < 0$

$\therefore$ a relative max T.P. at $h = \frac{2r^2}{L+2r}$
 $r = \frac{h^2}{L-h}$

MATHEMATICS Extension 1 : Question 7

Suggested Solutions

Marks

Marker's Comments

(c) (i) $(1-x)^{2n} = \binom{2n}{0} - \binom{2n}{1}x + \binom{2n}{2}x^2 - \binom{2n}{3}x^3 + \dots + \binom{2n}{2n}x^{2n}$ ①

(ii) By differentiating both sides wrt x
 $-2n(1-x)^{2n-1} = -\binom{2n}{1} + 2\binom{2n}{2}x - 3\binom{2n}{3}x^2 + \dots + 2n\binom{2n}{2n}x^{2n-1}$ ✓

Put $x=1$ ②
 $0 = -\binom{2n}{1} + 2\binom{2n}{2} - 3\binom{2n}{3} + \dots - (2n-1)\binom{2n}{2n-1} + 2n\binom{2n}{2n}$ ✓

so $\binom{2n}{1} + 3\binom{2n}{3} + \dots + (2n-1)\binom{2n}{2n-1} =$
 $= 2\binom{2n}{2} + 4\binom{2n}{4} + \dots + 2n\binom{2n}{2n}$ q.e.d.

1 For Diff eqn ...
 1 For subst $x=1$
 and